

Wrist pain

with GREGORY HOY and HAMISH ANDERSON*

Laura Robson admits she will have to start her career all over again after she recovers from major wrist surgery. Robson said she 'felt like a child again' in the aftermath of her operation, struggling to dress herself and cut her own food. Itching to return to action, Robson said she has been reduced to tears, at times frustrated with rehabilitation while her peers train and compete.

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The wrist is frequently injured during sport.¹ Distal radial fractures are common, as are scaphoid fractures and tendon problems. Carpal instabilities are less frequent but often career threatening.

Men are more likely to sustain a hand or wrist injury² and children/adolescents are more likely to have a wrist injury compared with adults.³ Injuries to the wrist range from acute traumatic fractures, such as occur during football, hockey and snowboarding, to overuse conditions, which occur more commonly in racquet sports, golf and gymnastics. If wrist injuries are not treated appropriately at the time of injury, they can lead to future impairments that may affect not only sporting endeavours, but also activities of daily living.⁴

In this chapter, we will assess wrist injuries from an anatomical viewpoint, based on the site of pain. They are:

- radial-sided wrist pain, both acute and chronic
- central wrist pain, including volar and dorsal
- ulnar-sided wrist pain.

The differential diagnoses of wrist pain are listed in Table 26.1.

CLINICAL APPROACH

The wrist joint has multiple axes of movement: flexion/extension and radial/ulnar deviation occur at the radio-carpal and midcarpal joints, and pronation/supination occurs at the distal and proximal radio-ulnar joints. These movements provide mobility for hand function.

The anatomy of the wrist is complex (Fig. 26.1). It is helpful to know the surface anatomy of the scaphoid

tubercle, hook of hamate, pisiform, Lister's tubercle and anatomical snuffbox. The bony anatomy consists of a proximal row (lunate, triquetrum, pisiform) and a distal row (trapezium, trapezoid, capitate, hamate), which are bridged by the scaphoid bone. Normally, the distal carpal row should be stable; thus, a ligamentous injury here can greatly impair the integrity of the wrist. The proximal row permits more intercarpal movement to allow wrist flexion/extension and radial and ulnar deviation. Here, a ligamentous injury disrupts important kinematics that have been recently labelled the 'dart thrower's motion' between the radius, scaphoid, lunate and the capitate in the central column, resulting in carpal instability with potential weakness and impairment of hand function. Recognising these ligamentous injuries early is not just important for athletic performance, but also for minimisation of any subsequent post-traumatic arthritis.

History

It is essential to determine the mechanism of the injury (if there is one) causing wrist pain. A fall on the outstretched hand may be severe enough to fracture the scaphoid or distal radius, or damage the intercarpal ligaments and/or triangular fibrocartilage complex. These injuries are commonly encountered in high-velocity activities such as snowboarding,⁵ rollerblading^{6, 7} or falling off a bike. A patient may fracture the hook of hamate while swinging a golf club,⁸ tennis racquet or bat and while striking a hard object (e.g. the ground). Rotational stress to the distal radio-ulnar joint and forced ulnar deviation and rotation may tear the triangular fibrocartilage complex.

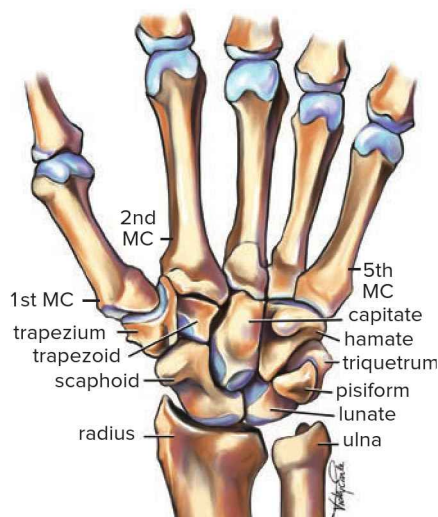
Table 26.1 Causes of wrist pain divided into regions

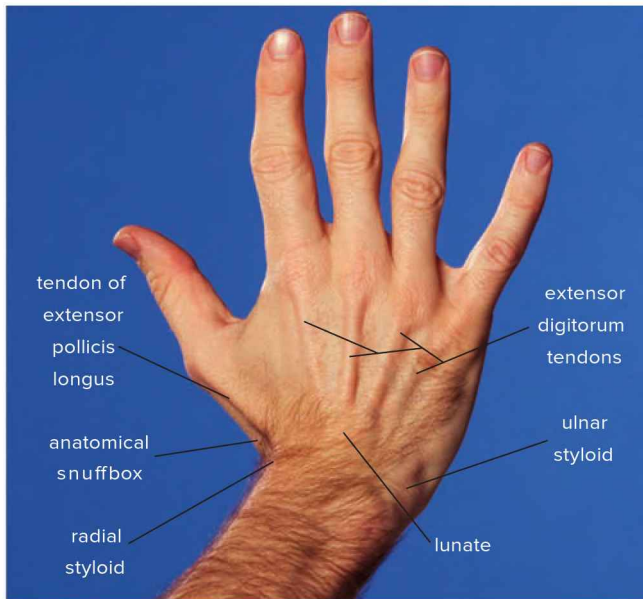
Zone	Common	Less common	Not to be missed
Dorsal wrist			
Radial zone	Extensor carpi radialis tenosynovitis De Quervain's tenosynovitis (volleyball) Scaphoid fracture Distal radius fracture (often intra-articular in the athlete)	Intersection syndrome (rowing/paddling) Anterior interosseous nerve compression Radial sensory nerve compression (volleyball) Extensor pollicis longus impingement on Lister's tubercle with occasional rupture (gymnastics)	Carpal dislocation Radial epiphyseal stress reaction (gymnastics)
Central zone	Posterior interosseous nerve/radial nerve entrapment Ganglion cyst Scapholunate ligament sprain (football)	Impingement syndromes: lunate (gymnastics); capitate (weightlifting); triquetrohamate (racquet sports); radial styloid (golf)	Scapholunate dissociation Perilunar dislocation Kienböck's disease
Ulnar zone	Extensor carpi ulnaris tenosynovitis (golf/tennis) Triangular fibrocartilage complex tear (water skiing) Ulnar styloid fracture Triquetral fracture (boxing)	Distal radioulnar joint instability Luno-triquetral ligament sprain Ulnar fracture Extensor carpi ulnaris subluxation (cricket)	Lunotriquetral dissociation
Volar wrist			
Radial zone	Carpometacarpal osteoarthritis	Flexor carpi radialis tendinopathy	Carpal instability
Central zone	Carpal tunnel syndrome (motorcycling)	Pisotriquetral degenerative joint disease	
Ulnar zone	Thoracic outlet syndrome Lunotriquetral ganglion	Hook of hamate fracture (golf) Ulnar nerve compression at Guyon's canal (cycling, baseball) Flexor carpi ulnaris tendinopathy	Anterior dislocation of lunate Traumatic ulnar artery aneurysm or thrombosis (karate)

When a patient presents with gradual-onset or chronic wrist pain, the clinician should consider whether the pain may be a manifestation of a systemic condition (e.g. metabolic disorder or spondyloarthopathy, see Chapter 7). Also, the clinician should rule out an uncommon presentation of radiating pain from a more proximal problem such as a herniated cervical disc.

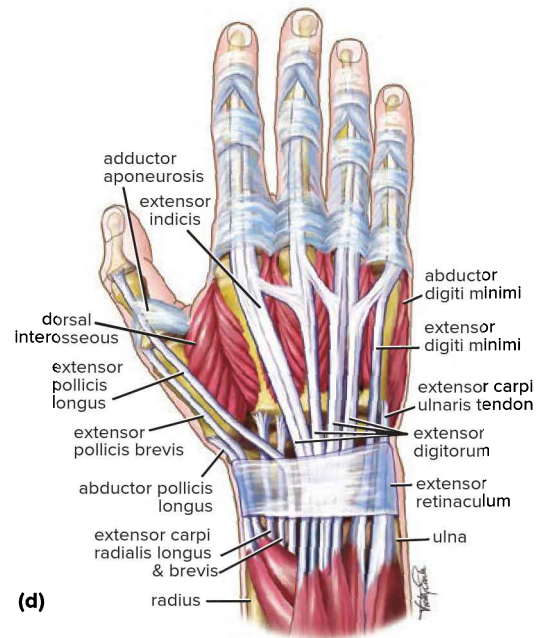
A detailed history will reveal whether the problem stems from an overuse condition (e.g. tenosynovitis), or from an acute injury that has not been correctly diagnosed or treated. Some patients suffer only minor discomfort at the time of an initial injury and fail to seek attention at that time. Factors that aggravate the pain provide useful information as to which structures are involved in chronic wrist pain.

Pain after repeated movement, with stiffness after a period of rest, suggests an inflammatory condition such as tenosynovitis. Pain aggravated by weight-bearing activities, such as gymnastics or diving, suggests bone or joint involvement. A history of joint clicking may be

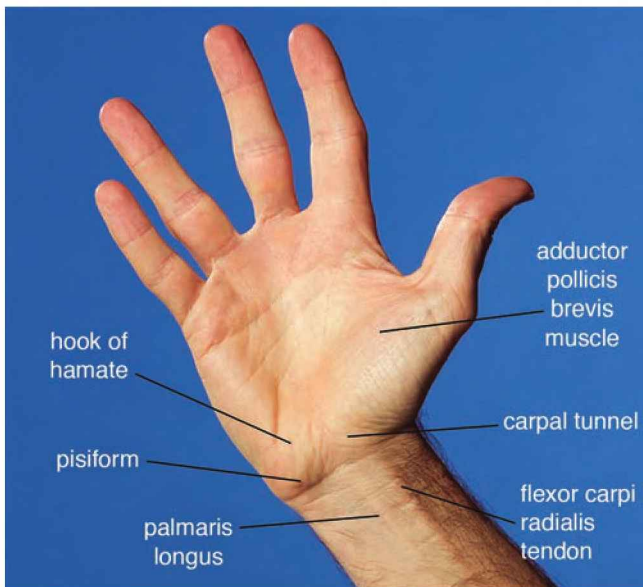
**(a)****Figure 26.1** Anatomy of the wrist (a) Carpal bones (MC = metacarpal)



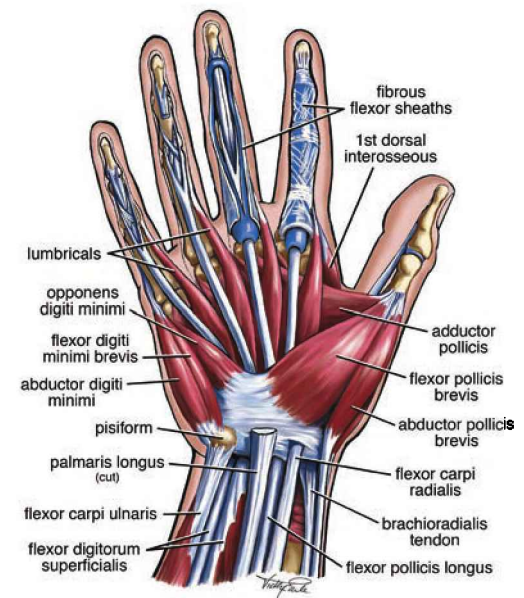
(b)



(d)



(c)



(e)

Figure 26.1 (cont.) (b) Surface anatomy, dorsal view (c) Surface anatomy, volar view (d) Dorsal aspect (e) Volar aspect

associated with carpal instability, triangular fibrocartilage tears or extensor carpi ulnaris subluxation. Characteristic night pain, with or without paraesthesia, is found in carpal tunnel syndrome. Associated neck or elbow symptoms suggest referred pain.

Other important aspects of the history may include:

- hand dominance
- occupation (e.g. computer-related, manual labour, food service industry)

- degree of reliance upon hands in occupation/recreation
- musician (number of years playing, hours of practice per week, change in playing, complex piece, etc.)
- gardening, crafts, hobbies.
- history of past upper extremity fractures including childhood fractures/injuries
- history of osteoarthritis, rheumatoid arthritis, thyroid dysfunction, diabetes
- any unusual sounds (e.g. clicks, clunks, snaps, etc.)
- recurrent wrist swelling raises the suspicion of wrist instability

Physical examination

Examination involves:

1. Observation (Fig. 26.2a)
Inspection may reveal a ganglion on the dorsum of the wrist. Swelling over the radial styloid may indicate de Quervain's tenosynovitis. Muscle wasting of the thenar or hypothenar eminence is found in the late stages of median or ulnar nerve compression, respectively.
2. Active movements
 - a. flexion/extension
 - b. supination/pronation
 - c. radial/ulnar deviation (Fig. 26.2b)
3. Passive movements
 - a. extension (Fig. 26.2c)
 - b. flexion (Fig. 26.2d)
4. Palpation
Palpate the wrist to detect tenderness and to determine whether the pathology appears to be extra-articular (i.e. soft tissue) or articular.
 - a. distal forearm (Fig. 26.2e)
 - b. radial snuffbox (Fig. 26.2f)
 - c. base of metacarpals
 - d. lunate (Fig. 26.2g)
 - e. head of ulna (Fig. 26.2h)
 - f. radio-ulnar joint
 - g. hamate/pisiform (Fig. 26.2i)
 - h. tuberosity of trapezium

On the volar aspect of the wrist, palpate the tuberosity of the trapezium as a bony prominence at the base of the thenar eminence.
5. Special tests
 - a. Watson's test for scapholunate injury (Fig. 26.2j)
 - b. stress of triangular fibrocartilage complex (Fig. 26.2k)
 - c. ulnar fovea sign for foveal disruption and ulnar triquetral ligament injury (Fig. 26.2l)
 - d. press test for triangular fibrocartilage complex pain (Fig. 26.2m)

- e. Finkelstein's test (Fig. 26.2n)
- f. Tinel's sign for nerve entrapment (Fig. 26.2o)
- g. Phalen's test for carpal tunnel syndrome
- h. two point discrimination (Fig. 26.2p).

The clinical utility of selected assessment findings is shown in Table 26.2.

Key outcome measures

Several valid and reliable assessment scales can quantify function of the wrist specifically or the upper extremity after an injury. These include the Patient Rated Wrist Evaluation (PRWE),^{15, 16} the Disability of the Arm, Shoulder and Hand (DASH and QuickDASH)^{17, 18} and the Mayo Wrist Score¹⁹ (Table 26.3).

Investigations

Plain radiography

Plain radiographs taken should include a posteroanterior (PA) view and a PA with both radial and ulnar deviation. If ligament injury is suspected, also obtain a PA view with clenched fist. A straight lateral, with the dorsum of the distal forearm and the third metacarpal forming a straight line, permits assessment of the distal radius, the lunate (or 'intercalated segment'), the scaphoid and the capitate, and may reveal subtle instability called DISI (dorsal intercalated segment instability) or VISI (volar intercalated segment instability). Undisplaced distal radial and scaphoid fractures, however, may be difficult to see on initial radiographs; clinical suspicion warrants use of other modalities (see 'Special imaging studies', below).

The normal PA view is shown in Figure 26.3a. Inspect each bone in turn. Note the line joining the proximal ends of the proximal row of the carpus and the C shape of the midcarpal joint (Gilula's arcs). If these lines are not smooth, a major abnormality is present. Assess the size of the scapholunate gap and look for scaphoid flexion (the signet ring sign) as these are signs of scapholunate instability.

The lateral radiograph of the normal wrist can be seen in Figure 26.3b. The proximal pole of the lunate fits into the concavity of the distal radius and the convex head of the capitate fits into the distal concavity of the lunate. These bones should be aligned with each other and with the base of the third metacarpal.

A clenched fist PA view should be taken if scapholunate instability is suspected. This is indicated by a widened gap of 3 mm or greater between the scaphoid and lunate on the PA view, but this may not present until some time after a scapholunate tear.



(a)



(b)

Figure 26.2 Examination of the patient with an acute wrist injury (a) Observation—inspect the wrist for obvious deformity suggesting a distal radial fracture. Swelling in the region of the radial snuffbox may indicate a scaphoid fracture. Inspect the hand and wrist posture, temperature, colour, muscular wasting, scars, normal arches of the hand (b) Active movement—radial/ulnar deviation. Normal range is radial 20° and ulnar 60° . Pain and restriction of movement should be noted. Always compare motion with that of the other hand (c) Range of motion—extension (the ‘prayer position’). Normal range of motion in wrist extension is 70° (d) Range of motion—flexion (the ‘reverse prayer position’). Normal range of motion in wrist flexion is $80\text{--}90^{\circ}$



(c)



(d)



(e)



(g)



(f)



(h)

Figure 26.2 (cont.) (e) Palpation—the distal forearm is palpated for bony tenderness or deformity (f) Palpation—radial snuffbox. The proximal snuffbox is the site of the radial styloid, the middle snuffbox is the site of the scaphoid bone, while the distal snuffbox is over the scaphotrapezoid joint (g) Palpation—the lunate is palpated as a bony prominence proximal to the capitate sulcus. Lunate tenderness may correspond to a fracture. On the radial side of the lunate lies the scapholunate joint, which may be tender in scapholunate ligament sprain. This is a site of ganglion formation. On the ulnar side of the lunate lies the triquetrolunate ligament. Tenderness and an associated click on radial and ulnar deviation of the wrist may occur with partial or complete tears of this ligament (h) Palpation—head of ulna and ulnar snuffbox. Swelling and tenderness over the dorsal ulnar aspect of the wrist is present with fractures of the ulnar styloid. Distal to the ulnar head is the ulnar snuffbox. The triquetrum lies in this sulcus and can be palpated with the wrist in radial deviation. Tenderness may indicate triquetral fracture or triquetrolunate injury. The triquetrohamate joint is located more distally. Pain here may represent triquetrohamate ligament injury

PRACTICE PEARL

Scapholunate instability cannot be excluded on a normal plain radiograph as the instability is initially dynamic, occurring only with grip.

Special imaging studies

The combination of the complex anatomy of the wrist and subtle wrist injuries that can cause substantial

morbidity has led to development of specialised wrist imaging techniques. Scaphoid views should be requested if a scaphoid fracture is suspected. A carpal tunnel view with the wrist in dorsiflexion allows inspection of the hook of hamate and the ridge of the trapezium. For suspected mechanical pathology, such as an occult ganglion, an occult fracture, non-union or bone necrosis, several modalities are useful (e.g. ultrasonography, radionuclide bone scan, computed tomography (CT) scan or magnetic resonance imaging (MRI) scan).



(i)



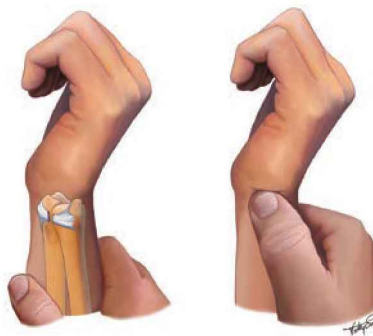
(k)



ulnar deviation

radial deviation

(j)



(l)

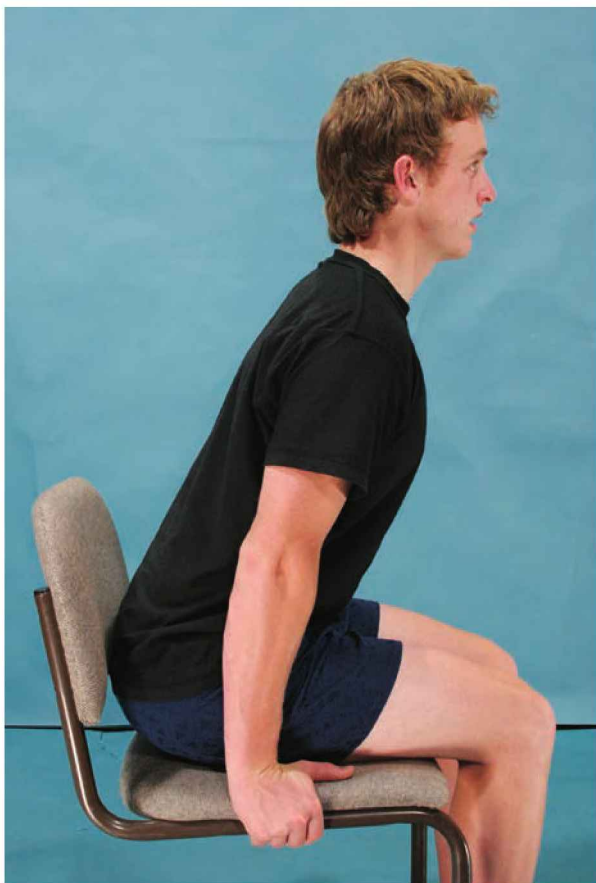
Figure 26.2 (cont.) (i) Palpation—the pisiform is palpated at the flexor crease of the wrist on the ulnar side. Tenderness in this region may occur with pisiform or triquetral fracture. The hook of hamate is 1 cm (0.5 in.) distal and radial to the pisiform. Examination may show tenderness over the hook or on the dorsal ulnar surface (j) Special test—Watson's test for scapholunate instability. The examiner places the thumb on the scaphoid tuberosity as shown, with the wrist in ulnar deviation. The wrist is then deviated radially with the examiner placing pressure on the scaphoid. If the athlete feels pain dorsally (over the scapholunate ligament) or the examiner feels the scaphoid move dorsally, then scapholunate dissociation is present (k) Special test—triangular fibrocartilage complex integrity. The wrist is placed into dorsiflexion and ulnar deviation and then rotated. Overpressure causes pain and occasionally clicking in patients with a tear of the triangular fibrocartilage complex (l) Special test—ulnar fovea sign for foveal disruption and ulnar triquetral ligament injury

Ultrasonography is a quick and accessible way to assess soft tissue abnormalities such as tendon injury, synovial thickening, ganglions and synovial cysts. Bone scans have high sensitivity but low specificity; thus, they can effectively rule out stress fractures, although MRI may be equally sensitive and more specific than a bone scan.

CT scanning is particularly useful for evaluating fractures that are difficult to assess fully on plain radiography, but MRI

can also provide information about soft tissue injuries. Thus, a complete scapholunate ligament tear is more effectively identified with MRI than with CT.

Arthrography of the wrist is no longer used as an investigative tool except in combination with MRI—MR arthrogram or MRA. If all imaging results are negative but clinically significant wrist pain persists, the clinician should refer the patient to a specialist for further evaluation.

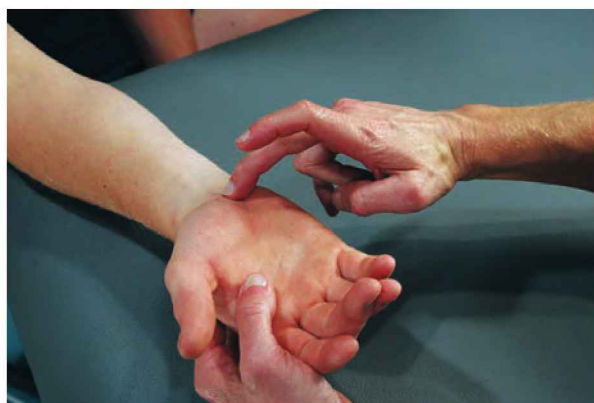


(m)

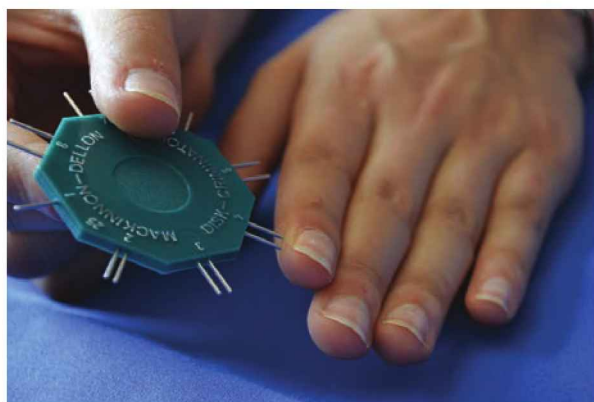
Figure 26.2 (cont.) (m) Special test—press test (or 'sitting hands' test). Attempting to raise body weight from a chair reproduces the pain of the triangular fibrocartilage complex injury (n) Special test—Finkelstein's test to detect de Quervain's disease. The thumb is placed in the palm of the hand with flexion of the metacarpophalangeal and interphalangeal joints while the examiner deviates the wrist in the ulnar direction (o) Special test—Tinel's sign. Tapping over the median nerve at the wrist produces tingling and altered sensation in the distribution of the median nerve in carpal tunnel syndrome (p) Special test—two point discrimination. Assessment of fine sensory ability. Apply the two points with even pressure horizontally over the medial and lateral aspects of the distal phalanx, starting wide and becoming narrower, with single point comparison



(n)



(o)



(p)

RADIAL COLUMN PRESENTATIONS

Fracture of the distal radius

Distal radius fractures are very common peripheral fractures.²³ As the force required to fracture young adults' bones is great, athletes may simultaneously incur an

intra-articular fracture and ligamentous strain or rupture. The higher the forces involved (e.g. in high-velocity sports), the greater the likelihood of a complex injury involving articular structures (Fig. 26.4). Thus, thorough assessment of ligamentous injury is essential when fractures occur.

Table 26.2 Clinical utility of selected assessment findings for wrist pain

Test	What is a positive finding?	Sensitivity	Specificity	LR+	LR–
Carpal tunnel syndrome					
Phalen's test ⁹	Reproduction of symptoms	0.92	0.88	2.68	0.54 ¹⁰
Tinel's sign ⁹	Paraesthesia in median nerve distribution or electric shock-like sensation	0.97	0.91	2.95	0.57 ¹⁰
Scapholunate injury					
Watson's scaphoid shift ¹¹	Reproduction of pain with or without hypermobility	69	66	1.78	0.55 ¹²
Triangular fibrocartilage complex disruption					
Ulna foveal sign ¹³	Exquisite tenderness that replicates pain	95.2	86.5	7.06	0.05
Press test ¹⁴	Axial ulnar load that replicates symptoms	100	NA	NA	NA

LR+ = positive likelihood ratio; LR– = negative likelihood ratio; NA = not available; see Chapter 14 for a guide to interpreting test sensitivity, specificity and likelihood ratios

Table 26.3 Patient-reported outcome measures for wrist pain

Patient Rated Wrist Evaluation (PRWE)
- Measures pain and disability in persons with wrist pathology with reference to functional activity 15 items - Score range: 0 (no impairment)–100 (worst score) - Test–retest reliability: >0.90¹⁶ - Validity established¹⁶ - MDC95% (minimal detectable change): 11.0²⁰ - MCID (minimal clinically important difference): 11.5²⁰ - Available online: www.physio-pedia.com/PRWE_Score
Disability of the Arm, Shoulder and Hand (DASH and QuickDASH) questionnaires
- Measure physical function, symptoms and quality of life in people with musculoskeletal disorders of the upper limb - DASH 30 items; QuickDASH 11 items - Score range: 0 (no impairment)–100 (worst score) - Test–retest reliability: DASH >0.96¹⁷; QuickDASH >0.91²¹ - Validity: DASH Pearson Realize >0.70¹⁷; QuickDASH Pearson Realize >0.70²² - MDC95%: DASH 12.75%¹⁷; QuickDASH 11.2%²¹ - MCID: 8%²¹ to 10.83%¹⁷ - Available online: www.physio-pedia.com/DASH_Outcome_Measure
Mayo Wrist Score
- Widely used but not validated, brief, objective and subjective assessment of pain/function/range of movement and strength 5 items - Score range: 0 (worst score)–100 (no impairment) - Available online: www.orthopaedicscore.com



(a)



(b)

Figure 26.3 Radiograph of the wrist (a) PA view—Gilula's arcs (b) Lateral view

Initial treatment of the fracture is anatomical reduction, which should be done even if later internal fixation is contemplated, thus reducing swelling and making later anatomical alignment more likely.

Conservative management by reduction and casting with radiological follow-up at 3 and 6 weeks is appropriate for stable fractures post-reduction with minimal risk of displacement. Casts should be moulded with the wrist flexed to no more than 20° and slight ulnar deviation, and permit full finger motion. As oedema reduces, the cast should be changed. There is no evidence to support the use of thermoplastic splints or braces for acute distal radius fracture.²⁴

Surgical intervention with early mobilisation for distal radius fracture has become the norm, with a particular emphasis on volar, locked plating. In spite of Cochrane reviews not being able to prioritise one surgical method over another, volar plating is being increasingly adopted.²⁵ Volar plating avoids the complications associated with external fixation and the surgical experience demands of fragment-specific fixation, as well as reducing the risk of extensor tendon rupture associated with dorsal plates.²⁵ Indications for open reduction are dorsal comminution, intra-articular involvement, instability on reduction, dorsal angulation of more than 20°, an articular surface step of greater than 1 mm, or radial shortening of greater than 5 mm, with anatomical alignment a necessity to avoid ongoing functional impairment.^{26, 27}

Post-operatively, there is little published evidence for early mobilisation with reference to long-term results;^{24, 28} however, a stable fixation, control of oedema and a controlled early mobilisation program in the athlete is crucial to an early return to sport, accompanied by suitable brace protection if appropriate. Inaccurate reduction, articular surface angulation, radial inclination or inadequate restoration of length all require early internal fixation with fixed angle-volar plating, although relative indications such as return-to-sport timelines and career goals are now just as important to the athlete.^{29, 30} Overall, there is a trend to more aggressive treatment using volar plating and this has led to improved functional outcomes, especially in the young active adult.

Fracture of the scaphoid

The most common carpal fracture involves the scaphoid⁴ and the usual mechanism is a fall on the outstretched hand. As pain may settle soon after the fall, presentation may be late after injury. The key examination finding is tenderness in the anatomical snuffbox. This may be accompanied by swelling and loss of grip strength. A more specific clinical test for scaphoid fracture is pain on axial compression of



(a)



(b)

Figure 26.4 Fracture of radius (a) Coronal CT of comminuted fracture of radius (b) Axial CT of comminuted fracture of radius

the thumb towards the radius or direct pressure on the scaphoid tuberosity with radial deviation of the wrist. Plain radiographs with special scaphoid views will usually demonstrate the fracture (Fig. 26.5a), but CT scan may be required to detect more subtle fractures (Fig. 26.5b).

Scaphoid fracture is the most commonly missed fracture leading to litigation. If there is no bony damage, scapholunate instability should also be considered (see below).

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If a scaphoid fracture is suspected clinically but the radiograph is normal, a fracture cannot be ruled out. CT is an ideal diagnostic test for an acute injury that may be cost-saving in some settings. Note that it can take 24 hours for the injury to be revealed on MRI.

Traditional treatment of stable and unstable scaphoid fractures

A stable scaphoid fracture (including stress fractures of the scaphoid, which occur in high-board divers and gymnasts³¹) may be immobilised for eight weeks in



(a)

Figure 26.5 Scaphoid fracture (a) Plain radiograph of subtle scaphoid fracture (arrow)



(b)

Figure 26.5 (cont.) (b) CT requested in plane of scaphoid shows scaphoid fracture (arrow)

a scaphoid cast extending from the proximal forearm to, but not including, the interphalangeal joint of the thumb (Fig. 26.6a). Upon removing the cast, re-evaluate the fracture clinically and radiologically. Unstable, angulated ($>15\text{--}20^\circ$) or significantly displaced fractures (diastasis in the fracture gap $>1.5\text{ mm}$) require immediate percutaneous fixation (Fig. 26.6b) or open reduction and internal fixation. There are initial reports of using scaphoid plates but no long-term follow-up.

PRACTICE PEARL

Current treatment of scaphoid fractures

The most reliable method of achieving scaphoid union with anatomic position is with internal fixation using mixed threaded (or graduated thread) screws which compress the scaphoid.

Complications of scaphoid fracture

The blood supply to the scaphoid originates distally (Fig. 26.7a), so the flow to the proximal pole can be diminished, which can then be at risk of necrosis after a fracture (Fig. 26.7b). Scaphoid fractures also have a risk of delayed union or non-union. Non-union and



(a)

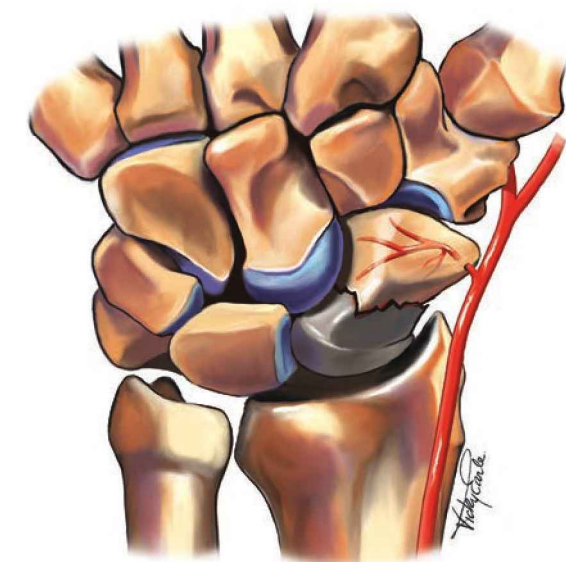


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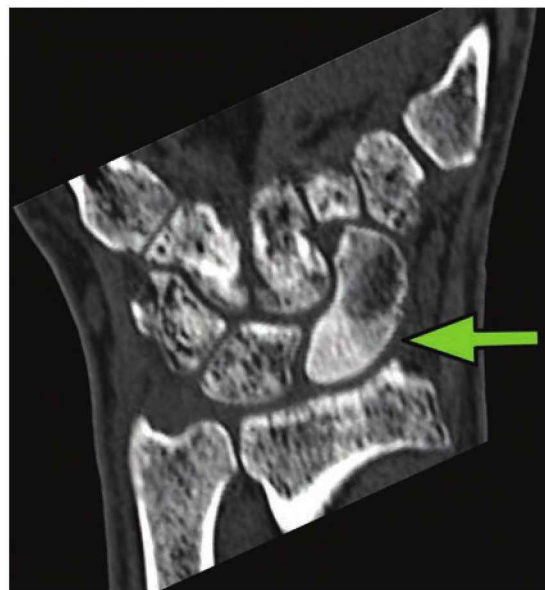
Figure 26.6 Treatment of scaphoid fractures (a) Cast immobilisation (b) Surgical fixation is now first-line treatment for scaphoid fracture

osteonecrosis can occur on their own or can occur in combination.³²

If there is clinical evidence of incomplete union when the cast is removed, the fracture should be immobilised for a further four to six weeks. Further immobilisation beyond this time is unlikely to prove beneficial. CT scan is the investigation of choice to detect non-union, but MRI can be used if CT is not available.



(a)



(b)

Figure 26.7 (a) The artery to the scaphoid enters via the distal pole, so a fracture at the waist of the scaphoid compromises the proximal fragment (b) MRI showing the characteristic hypointense appearance of avascular necrosis (arrow) in the proximal pole of the scaphoid

Contemporary management of scaphoid non-union

Contemporary treatment of non-union requires specialist hand surgeon management. Simple non-unions may be treated with fixation with bone graft (if there is a deformity or there has been bone resorption), while osteonecrosis with or without non-union may require a vascularised bone graft.³³

Post-immobilisation wrist rehabilitation

Following successful treatment of a scaphoid fracture, the athlete can gradually return to activity using a protective device. Compression tubing worn under the protective splint reduces oedema and improves comfort. Different sports have different rules about what constitutes an 'allowable' protective cast (Fig. 26.8).

Fracture of the trapezium

Fractures of the trapezium make up 1-5% of carpal fractures. The most common pattern is a vertical, intra-articular fracture associated with axial compression from the thumb metacarpal. Rarely occurring in isolation, it often accompanies a Bennett's fracture. Presentation will be point tenderness and localised swelling, with the fracture itself best confirmed on CT scan (Fig. 26.9). Examination should be thorough because of the high



Figure 26.8 Following a scaphoid fracture, the athlete may wear a protective cast

risk of associated fracture dislocation and/or fractures of neighbouring carpals.³⁴

Intervention will be determined by the amount of displacement of the fracture. The risk of degenerative arthritis at the trapezial articulations is high for a displaced fracture in a young, active individual, and can lead to reduced carpometacarpal motion, with a painful pinch and grip. Surgery can involve percutaneous pinning or open reduction with screw fixation. Non-displaced fractures do well with thumb spica casting.



Figure 26.9 CT scan of fracture of trapezium

Radial epiphyseal injury (gymnast's wrist)

Skeletally immature athletes may place substantial compressive and shear forces across the distal radial physes, making them prone to growth arrest. Gymnasts, both elite and recreational, are particularly at risk of this injury, which occurs due to repetitive loading through a dorsiflexed wrist. It has been frequently hypothesised that this injury then puts them at risk of developing a positive ulnar variance; however, no definitive relationship has been established.³⁵

The patient will present with dorsal wrist pain that is directly associated with activity. Radiographic changes in order of severity are haziness around the physis, cystic changes and/or metaphyseal sclerosis and widening of the physis itself (Fig. 26.10). Salter Harris V stress fracture should be considered if the growth plate is narrowed. The adolescent (ages 10–14) is more at risk than the younger athlete.³⁶

Recovery can take several months and is activity dependent. Both athlete and family need to be aware of the long-term consequences of poor management. Treatment includes rest, adaptation of technique and forearm strengthening, particularly the flexors. Bracing to reduce hyperextension of the wrist has been shown in biomechanical studies to reduce load, and should be considered along with alternating swinging and loading skills during training.³⁷

De Quervain's tenosynovitis

De Quervain's tenosynovitis affects the synovium of the abductor pollicis longus and extensor pollicis brevis

tendons as they pass in their synovial sheath in a fibro-osseous tunnel at the level of the radial styloid (Fig. 26.11). This is a common radial-sided tendinopathy in athletes and occurs particularly with racquet sports, ten-pin bowlers, rowers and canoeists.



(a)



(b)

Figure 26.10 Radial epiphyseal injury in gymnast showing widening of the radial growth plate indicating stress reaction (a) Anteroposterior wrist X-ray (b) Lateral wrist X-ray

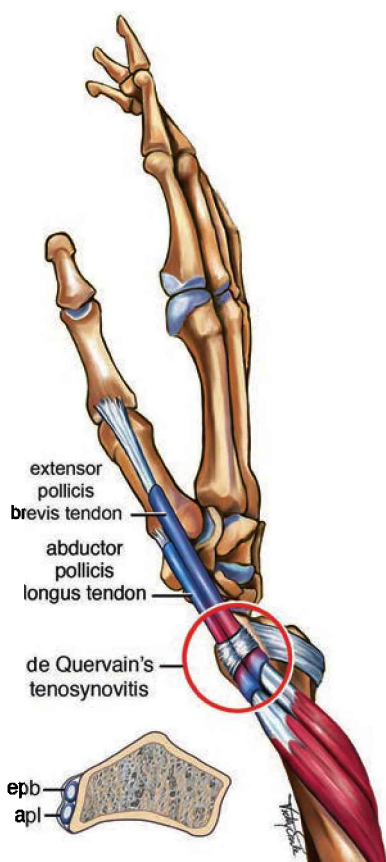


Figure 26.11 Site of compression in de Quervain's tenosynovitis

The left thumb of a right-handed golfer is particularly at risk because of the hyperabduction required during a golf swing. There will be local tenderness and swelling, which may extend proximally and distally along the course of the tendons. In severe cases, crepitus may be felt. A positive Finkelstein's test (Fig. 26.2n) is suggestive of the diagnosis; however, flexor carpi radialis tendinopathy also causes a positive test.

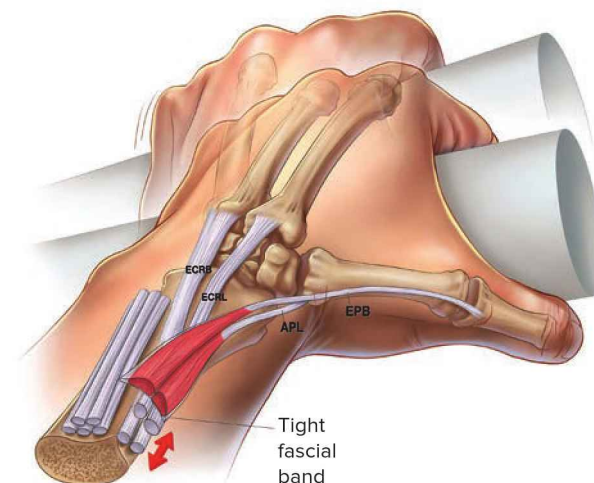
Treatment includes splinting, local electrotherapeutic modalities, stretches and graduated strengthening. Patients often find a pen build up (a rubber addition to enlarge the diameter of the pen) or golf grip widener useful as this reduces the stretch on the extensor tendons. An injection of corticosteroid and local anaesthetic into the tendon sheath will usually prove helpful.³⁸ In rare cases, surgical release is necessary.

Intersection syndrome

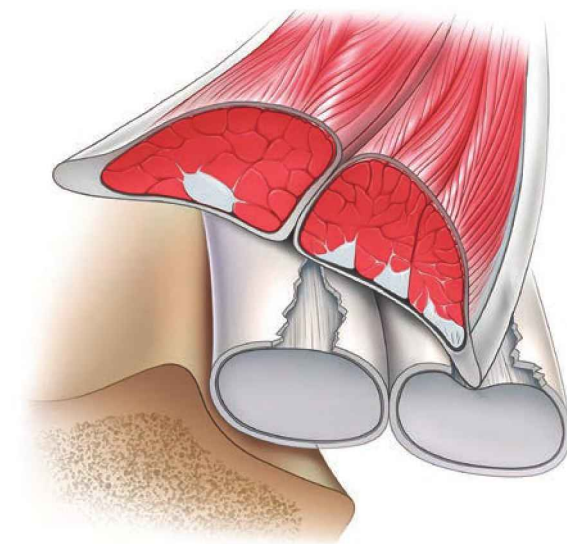
Intersection syndrome is a true tendinitis that occurs between the first and second dorsal wrist tendon compartments (Fig. 26.12), just proximal to the extensor retinaculum. It may be due to friction at the site of crossing or it may also

arise from compartment syndrome of the thumb extensors, causing a tight band over the wrist extensors. Tenderness is found dorsally on the radial side, with swelling and crepitus a short distance proximal to the site of maximal tenderness in de Quervain's disease (Fig. 26.11). This condition is sometimes called 'oarsman's wrist' because of its common occurrence in rowers,³⁹ but it is also seen in canoeists and in weight training and racquet sports.

Treatment involves early intervention with corticosteroid injection into the junction. Acute surgical



(a)



(b)

Figure 26.12 Intersection syndrome (a) Tight fascial band rubbing on tendons COURTESY OF DR GREG HOY (b) Close up of pathology COURTESY OF DR GREG HOY

decompression⁴⁰ and immediate return to training and events for rowers is very effective, and removes the need for changing rowing technique.³⁹

Radial sensory nerve compression

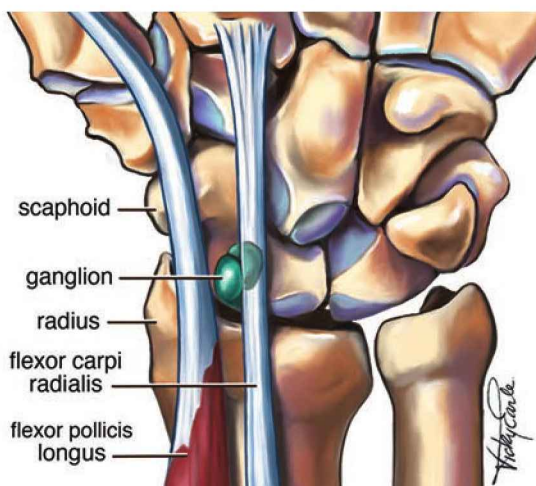
As the dorsal radial sensory nerve exits brachioradialis 5–7 cm proximal to the wrist, it becomes superficial and subject to damage. Compression can occur following radial styloid fracture, be caused by a tight cast or by repetitive trauma in sports like volleyball. It can also be compressed more proximally when squeezed between brachioradialis and extensor carpi radialis longus tendons during rapid, repeated pronation and supination.

It is differentiated from de Quervain's tenosynovitis by the presence of sensory changes over the radiodorsal aspect of the hand. Treatment is initially conservative with corticosteroid injection, splinting and addressing of potential causes of entrapment. Surgical decompression has good results in chronic conditions.⁴¹

CENTRAL COLUMN PRESENTATIONS

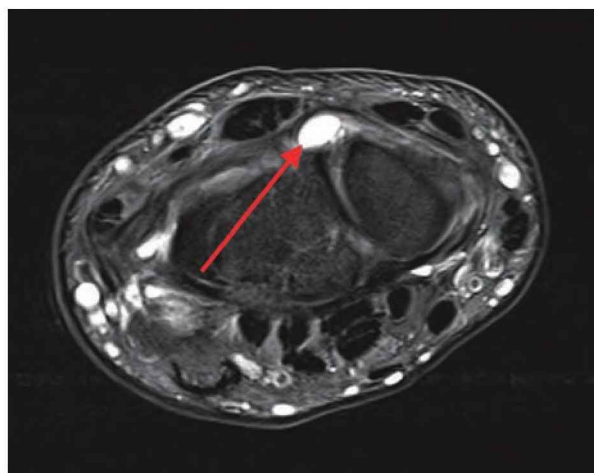
Ganglions

Ganglions occur in athletes of any age. They are a synovial cyst communicating with the joint space (Fig. 26.13a). They most often present as a relatively painless swelling. They occur in several common sites on both the dorsal and volar aspects of the wrist, most commonly the scapholunate space, due to the poorly named 'mucinous degeneration' of the scapholunate ligament. They may also be intracapsular or even intra-osseous.

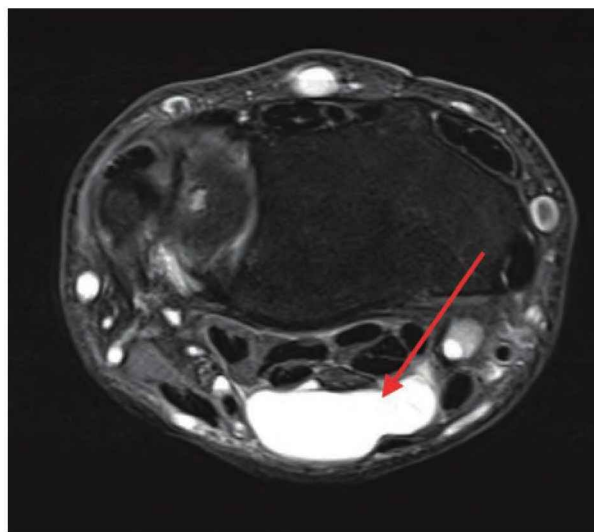


(a)

Figure 26.13 Ganglion cyst (a) Graphic view of the dorsum of the wrist showing the ganglion arising from the joint



(b)



(c)

Figure 26.13 (cont.) (b) MRI axial view of dorsal scapholunate ganglion (c) MRI axial view of volar wrist ganglion

The patient's main complaint is of intermittent wrist pain and reduced movement. Swelling may be visible intermittently or not at all, and so should not be relied upon to make the diagnosis. Ultrasonography is a useful investigation, however, T2-weighted MRI highlights ganglion cysts (Fig. 26.13b & c) and is the investigation of choice. The athlete must be reassured that the ganglion is benign, and that the average number in normal people is approximately two per wrist.⁴²

Treatment is only indicated for a symptomatic ganglion. When symptoms persist, aspiration and/or

corticosteroid infiltration are at least temporarily effective and can be performed under ultrasound guidance where this is feasible. Some persistent symptomatic ganglions require surgery, in which the neck of the ganglion must be removed to prevent recurrence.

Dislocation of the carpal bones

There are a number of different types of dislocation of the carpal bones, mostly involving the lunate. They are the uncommon end stage of severe ligament disruption. Failure to recognise them generally results in disastrous consequences.

Anterior dislocation of the lunate

In rare cases, the lunate may dislocate anteriorly as the end point of a perilunar injury. Pain is usually severe and deformity obvious. Plain radiography reveals the dislocation best in the lateral view with the lunate tilted volarly and not articulating with the capitate.⁴³ Treatment is URGENT open reduction and primary ligament repair, followed by eight weeks of cast immobilisation. Anterior lunate dislocation is often associated with severe median nerve compression and numbness in the radial three and a half digits.⁴⁴ This requires urgent surgical decompression.

Perilunar dislocation of the lunate

Perilunar dislocation is occasionally associated with a fractured waist of the scaphoid (so-called 'trans-scaphoid perilunate dislocation') when the lunate remains with the radius and the capitate is dislocated dorsally. This complex injury is often overlooked if the deformity is moderate, but the sequelae are catastrophic for the wrist, and this is a common cause of litigation when missed. It requires treatment by a hand and wrist surgeon, as long-term instability and radiographic wrist arthritis can occur.^{45, 46}

Scapholunate dissociation

Scapholunate dissociation is due to a complex injury involving dorsal scapholunate ligament tear and loss of secondary restraints (Fig. 26.14a). Rotatory subluxation (mainly flexion) of the scaphoid may occur as a result of disruption of its ligamentous attachments due to acute trauma (e.g. a fall on the dorsiflexed hand). Examination reveals tenderness 2 cm distal to Lister's tubercle on the radial side of the lunate. There may be little or no swelling. The key examination manoeuvre is Watson's test (Fig. 26.2j). If the test causes pain or reveals dorsal movement of the scaphoid, scapholunate instability is likely.

Conventional radiographic views may not show any abnormality, but stress films (clenched fist postero-anterior view) may reveal a scapholunate gap (Fig. 26.14b). A lateral radiograph may show an increased volar flexion of the distal pole of the scaphoid and dorsiflexion of

the lunate. If these tests are negative but the injury is suspected clinically, MRI is indicated (Fig. 26.14c).

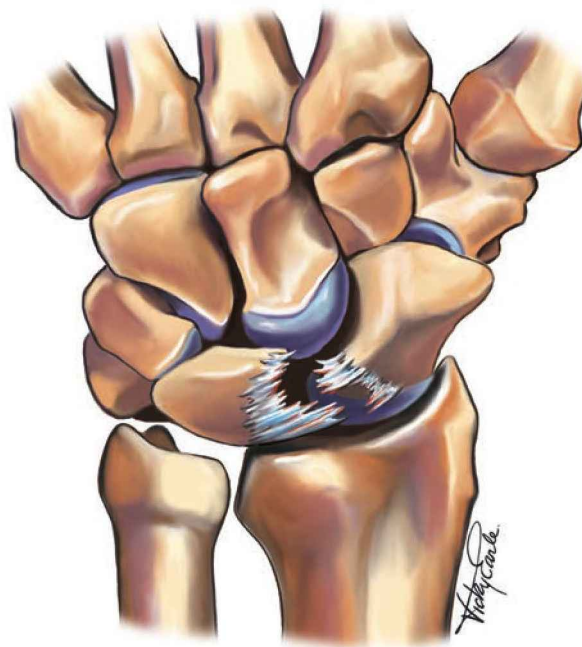
Treatment of scapholunate dissociation is open reduction and reconstruction of the ligaments with temporary internal fixation. Many techniques are published^{47, 48} and the main aim in athletes is return of grip strength and pain relief. Prevention of future arthritis (scapholunate advanced collapse (SLAC) wrist) is the ideal, although there is currently no significant evidence that this is achieved.

Kienböck's disease

Kienböck's disease is otherwise known as avascular necrosis of the lunate. It is associated with repetitive microtrauma and negative ulnar variance, and it is most commonly diagnosed in adult males. However, there is a published case report of it occurring in a 14-year-old female gymnast.⁴⁹

The patient will report a gradually worsening history of wrist pain. There may be swelling over the lunate which will be painful to palpate, and wrist motion will be affected. Imaging confirmation is increasingly using MRI. Interpretation of what is currently classified as Lichtman's stage 1 on MRI is controversial, with some authors arguing that this is an example of lunate overload and not Kienböck's disease.⁵⁰

Surgical treatment will depend on the stage of the disease and the patient's level of function. Radial



(a)

Figure 26.14 Scapholunate dissociation (a) Coronal graphic shows a tear of the dorsal component of the scapholunate ligament



(b)



(c)

Figure 26.14 (cont.) (b) In a classic case, radiograph reveals separation of the scaphoid and lunate. This can be a late sign, so normal radiography does not rule out this condition (c) Coronal MRI T2 image of scapholunate instability (arrow)

shortening osteotomy has been shown to have had good long-term results, with only slight loss of grip strength.⁵¹ Salvage procedures are required for advanced cases due to the presence of carpal collapse.⁵²

Conservative management of prolonged casting with consistent radiological follow-up has been shown not to be sufficient to stop progression of the disease once accurately diagnosed. Reported success of casting is possibly an indication that what was thought to be early-stage Kienböck's was in fact something else.⁵³ For this reason, it is important to differentiate between a stress reaction of the lunate and Kienböck's disease. Tennis players and other high-use athletes presenting with overuse syndromes in the wrist require exclusion of bony stress reactions including MRI scanning of good quality.

Diagnostic criteria for MRI for stage 1 Kienböck's disease have not been defined in the literature. We do not consider the use of proton density (PD) and T2 MRI images alone as sufficiently reliable for the diagnosis of Kienböck's disease. We recommend that T1 images must be obtained to exclude Kienböck's disease, and stress reactions must be considered in the differential diagnosis of these patients (Fig. 26.15).

Conservative treatment with restriction of high-level impact loading is excellent for the relief of these bony



Figure 26.15 MRI coronal view of Kienböck's disease of lunate

stress injuries when diagnosed in the first three months of symptoms. Alteration of technique deficits may prevent recurrence, but little is known of the possible cause-and-effect relationship with completed fractures. The evidence base regarding treatment of this condition is controversial.⁵⁴

Impingement syndromes

A number of impingement syndromes may cause wrist pain. Scaphoid impaction syndrome may occur because of repetitive hyperextension stresses (e.g. in weightlifting or gymnastics). This mechanism is also responsible for avascular necrosis of the capitate in weightlifters. Impaction of the dorsal pole of the lunate on the distal radius is seen in gymnasts. The extensor pollicis longus may impinge on Lister's tubercle and it occasionally ruptures. Triquetrohamate impingement syndrome may result from forced wrist extension and ulnar deviation (e.g. in racquet sports and gymnastics).

Radial styloid impaction syndrome can result from repeated forced radial deviation, especially among golfers (Chapter 9). Patients with these syndromes present with localised tenderness and are treated with rest and a protective brace. Occasionally, corticosteroid injection or surgical exploration may be helpful.

ULNAR COLUMN PRESENTATIONS

Ulnar styloid fracture

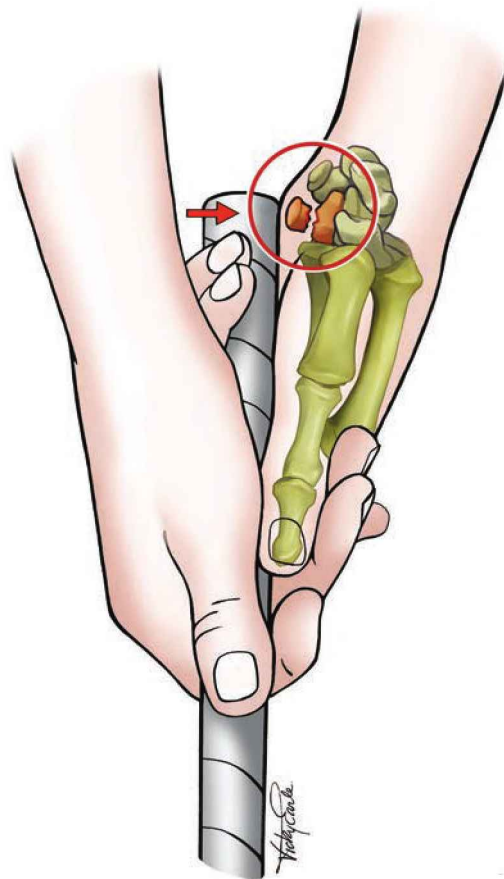
Ulnar styloid fractures are present in over 50% of distal radius fractures, with 25% of them never uniting. There is no benefit in fixation of an ulnar styloid fracture unless it can be demonstrated that it is associated with distal radio-ulnar joint (DRUJ) instability.⁵⁵ Its presence should alert the clinician to assess for triangular fibrocartilage complex injury and monitor for ulnar instability through the rehabilitative period. If the fracture requires addressing, options include small k-wires with a tension band or, when DRUJ instability is present, volar plating of the radius and supination splinting.⁵⁶ A recent study did demonstrate that patients with distal radius fractures who had an associated ulnar styloid fracture had decreased function and reported more pain initially; however, by 12 months these differences were no longer significant.⁵⁷

Fracture of the hook of hamate

Fracture of the hook of hamate may occur while swinging a golf club,^{8, 58, 59} tennis racquet or baseball bat. The fracture is especially likely to occur when the golf club strikes the ground instead of the ball, forcing the top of the handle of the club against the hook of the hamate of the top hand (Fig. 26.16a). This mechanism may compress

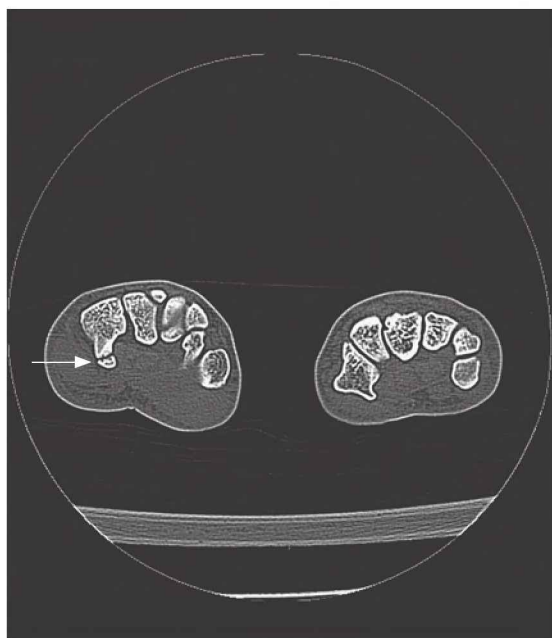
the superficial and deep terminal branches of the ulnar nerve, producing both sensory and motor changes. Symptoms include reduced grip strength and ulnar wrist pain. Examination reveals volar wrist tenderness over the hook of hamate. Routine radiographs of the wrist do not image the fracture and even the classic 'carpal tunnel view' with the wrist in dorsiflexion is an insensitive test. CT scan (Fig. 26.16b) and MRI are the best imaging tools.

This fracture often fails to heal with immobilisation; most sports medicine cases of the fracture are actually stress fractures that present late. In some athletes (e.g. baseball players), the hook of hamate fracture is likely to be a completed stress fracture not due to acute trauma. If diagnosis is delayed, or the fracture fails to heal clinically within four weeks of immobilisation, current surgical



(a)

Figure 26.16 Fracture of the hook of hamate (a) Possible mechanism—when the golf club is suddenly decelerated (e.g. hitting the ground), the grip is forced against the hook of hamate



(b)

Figure 26.16 (cont.) (b) CT scan of a fracture of the hook of hamate

practice is excision of the fractured hook followed by three weeks wrist immobilisation (in preference to open reduction and internal fixation).^{8,60} The pain only dissipates slowly, but the patient can usually resume sport six weeks after surgery. The long-term outcome of this approach is not known.

Triquetral fracture

Fractures of the triquetrum are the second most common carpal fracture. The two primary injury patterns are a fracture of the triquetral body, or the cortical, dorsal flake fracture which accounts for 93% of all triquetral fractures.

One mechanism of injury for the flake fracture is the ulna chiselling an avulsion fragment following a fall into forced ulnar deviation and dorsiflexion. A less common cause is a high-velocity dorsal blow forcing radial deviation and palmar flexion.⁶¹

The athlete will present with point tenderness over the dorsum and a painful wrist with flexion. Precise palpation of the fracture is difficult due to its ulnar position. Diagnosis is then dependent upon anteroposterior, lateral and oblique pronated view radiographic scans (Fig. 26.17). CT is a useful tool for occult fractures.

With respect to the flake fractures, the athlete must understand that it is not the bone that is the problem, but rather the soft tissue injury and potential loss of



Figure 26.17 Lateral radiograph of triquetral flake fracture (arrow)

stability. Cast immobilisation is essential for three to four weeks to ward off instability. Graded range of motion can begin then with a removable splint for protection, with full function usually restored by eight weeks.⁶²

The less common triquetral fracture is through the body of the bone following a high-energy collision. Any diagnosis of a fracture must be investigated thoroughly with 12-25% of these associated with perilunate fracture dislocation. Surgery using pins or a compression screw will depend on the degree of displacement and the diagnosed presence of concomitant injury.

Lunotriquetral dissociation

Lunotriquetral dissociation is an injury that is frequently missed on initial presentation, yet is the second most common form of carpal instability. The injury is often the result of a backwards fall onto an externally rotated, dorsiflexed, supinated and radially deviated hand. Physical examination is unlikely to reveal significant bruising or swelling. There will be pain at end range of pronation and supination and with pronated grip. Ballotement of the joint may be uncomfortable, and relief may be experienced with posterior displacement of the pisiform.⁶³

Conservative management can be successful in managing this condition and is preferable to surgery at least in acute settings. The wrist should be splinted or casted for six weeks, at which time a weaning and

ulnar-sided strengthening program can begin. The use of a pisiform boost type of splint may be beneficial in chronic cases. Athletes should be aware that the process of recovery from either conservative or surgical management can take up to six months.^{63, 64} Biomechanical studies indicate that training of extensor carpi ulnaris (ECU) may be beneficial; however, this is yet to be demonstrated clinically.⁶⁵

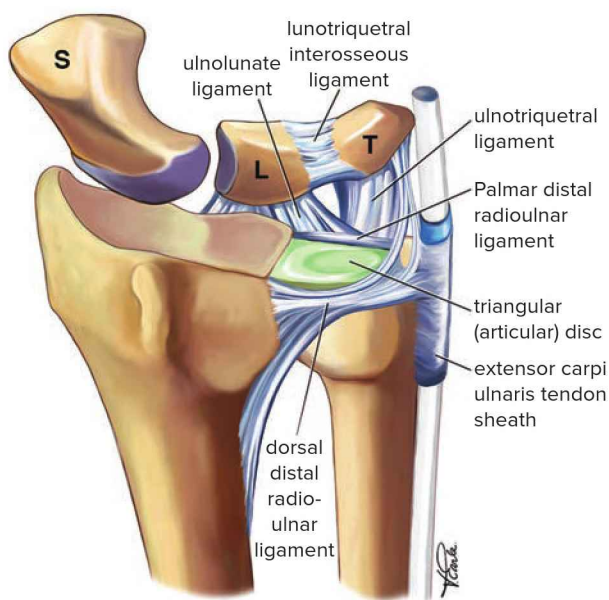
Triangular fibrocartilage complex tear

The triangular fibrocartilage complex (TFCC) lies between the ulna and the carpus. It is the major stabiliser of the distal radioulnar joint. The 'complex' consists of the triangular fibrocartilage, ulnar meniscus homologue, ulnar collateral ligament, numerous carpal ligaments and the ECU tendon sheath (Fig. 26.18a). Only the periphery of the TFCC is vascular and able to heal if damaged. The central avascular portion cannot heal and usually requires debridement once injured.

The TFCC is commonly injured in racquet or stick sports and those that create repetitive ulnar and compressive loading like gymnastics, weightlifting, boxing and surfing. In water skiers it is considered a traction injury.¹ It should be investigated following distal radius fracture and wherever DRUJ instability exists. The patient complains of ulnar-sided pain that worsens with grip, rotation and weight-bearing.⁶⁶

Examination reveals tenderness and swelling over the dorsal ulnar aspect of the wrist, pain on resisted wrist dorsiflexion and ulnar deviation (Fig. 26.2k), a clicking sensation on wrist movement and reduced grip strength. Sharp pain with deep palpation is known as a positive foveal sign and indicates an ulnatriquetral ligament injury and/or a disruption of the TFCC at the base of the ulna styloid (fovea), which is associated with DRUJ instability (Fig. 26.2l).^{13, 66} The 'press test' may also be helpful in diagnosing TFCC tears and differentiating from other conditions like ECU tendinopathy (Fig. 26.2m). The patient creates an axial ulnar load by attempting to lift his or her weight up off a chair using the affected wrist. A positive test replicates the presenting symptom.¹⁴

High-quality MRI (Fig. 26.18b) can image the TFCC and this is an increasingly popular investigation for ulnar-sided wrist pain.⁶⁷ Estimates of sensitivity and specificity are about 60% and 90%, respectively,⁶⁸ which suggests that a negative MRI should not be used to rule out the condition if it is clinically suspected. Interestingly, ultrasonography shows promise for matching MRI in the detection of TFCC lesions; however, arthroscopy remains the gold standard and should be considered as early as possible for the athlete wanting a rapid return to play.^{69, 70}



(a)



(b)

Figure 26.18 Triangular fibrocartilage complex (TFCC) (a) TFCC and surrounding ligaments (b) Coronal MRI of foveal tear of TFCC off ulna

Arthroscopy permits accurate diagnosis and excision of any torn cartilage if required. If the ulna is longer than the radius (positive ulnar variance), it impinges on the triangular fibrocartilage and predisposes it to tearing.

It may be necessary to shorten the ulna as well as excising the torn fibrocartilage.⁷¹

Treatment after surgery will likely involve a period of immobilisation, followed by a carefully graded return to full motion and grip strength.⁶⁶ Conservative management involves a short period of splinting followed by specific strapping to offload the TFCC, coupled with stabilisation exercises involving ECU and pronator quadratus, and likely technical adjustment. It is rarely as successful for treatment of central and foveal tears as it is for smaller, peripheral tears.

Distal radio-ulnar joint instability

The thickened dorsal and volar aspects of the triangular fibrocartilage act as the dorsal and volar ligaments of the distal radio-ulnar joint. Subluxation of the ulnar head occurs because of avulsion of these ligaments. It may be either volar or dorsal. Dorsal subluxation of the ulnar head associated with a tear of the volar radio-ulnar ligament is more common and may be due to repetitive or forceful pronation in contact sports, tennis or gymnastics. Dorsal displacement of the ulnar styloid process during pronation may be detected on true lateral radiograph. Treatment requires repair of the TFCC.⁷²

Extensor carpi ulnaris tendon injuries

The ECU resides in the sixth dorsal wrist compartment and helps stabilise the DRUJ.^{65, 73} It has its own subsheath just distal to the ulnar styloid, which prevents subluxation from the distal ulnar groove. Its angular path from its origin on the lateral epicondyle to the tendon insertion at the base of the fifth metacarpal is exaggerated with ulnar deviation and supination (Fig. 26.19a).

The subsheath is designed in such a way that it stretches and deforms in response to forces. It can fail abruptly with hypersupination and flexion, but more usually fails as fibrous tearing occurs over a period of time due to repetitive forearm supination, wrist flexion and ulnar deviation.

The tendon can either sublux (a partial loss of excursion beyond the volar lip of the notch, which occurs in up to 45% of asymptomatic persons) or it can dislocate with a complete loss of continuity. Presentation will be ulnar-sided wrist pain that intensifies with active contraction or stretching of the tendon. Instability will be obvious as the tendon subluxes volarly, often with an audible flick, and ultrasound imaging is useful.^{74, 75} Weight-bearing through the palm should not be painful unless other structures like the TFCC are involved.

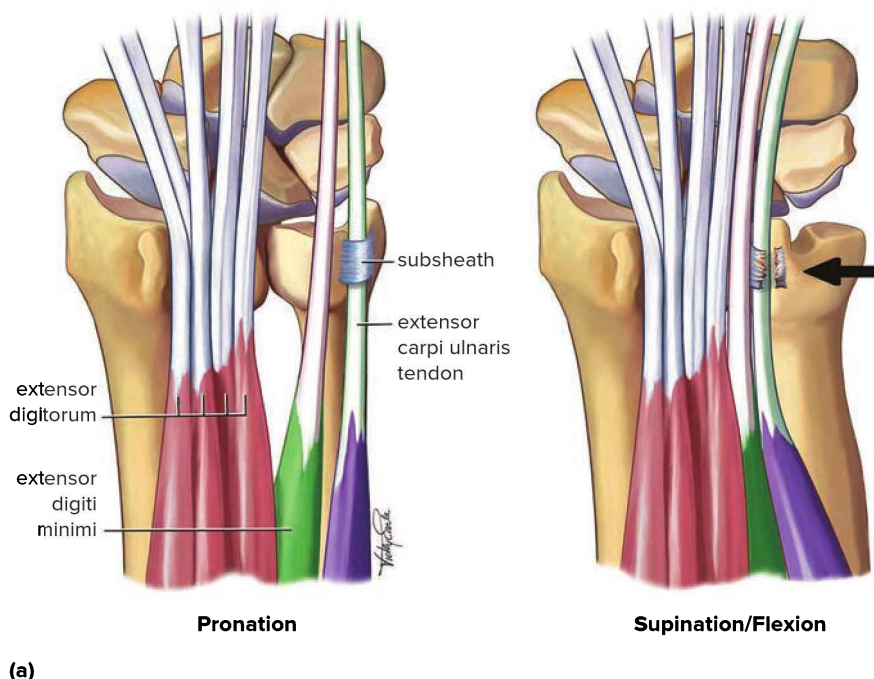
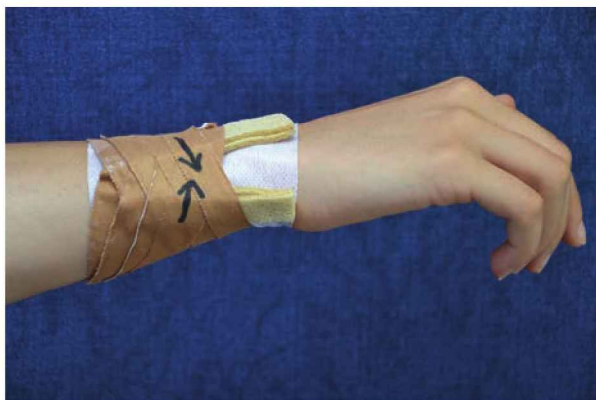


Figure 26.19 Extensor carpi ulnaris (ECU) tendon (a) Tendons at the wrist showing supination position where the ECU tendon is most likely to sublux and tear subsheath



(b)

Figure 26.19 (cont.) (b) Tennis-specific taping method that unloads and blocks the wrist with TFCC and/or ECU injury. Readers are directed to Stroia et al.⁷⁸ for a complete description of the method

Athletes at risk are those who require grip and extremes of wrist motion in supination, for example golfers, cricketers, weightlifters and rodeo riders.⁷⁶ Tennis players with double-handed backhands are at particular risk as the hand closest to the racquet head moves rapidly into a supinated, flexed and radially deviated posture.⁷⁷

If there is no tendon rupture, initial treatment can be conservative. The arm is splinted in radial deviation, extension and neutral to pronation. If stability remains a problem, the cast should extend to include the elbow at 90°. ⁷⁵ The splint is worn for 1–4 weeks, depending on the severity of the symptoms. A strengthening and control program is then initiated with taping to stabilise the ECU within the ulnar groove (Fig. 26.19b).⁷⁸ Failure of conservative management is an indication for surgical reconstruction of the sheath with or without recontouring of the sulcus. Anticipated return to play is at three months following surgery.⁷⁹

Other tendinopathies around the wrist

Any of the flexor and extensor tendons around the wrist may become painful with excessive activity. On examination, there is tenderness and occasionally swelling and crepitus. The principles of treating tendinopathies apply—management should include attention to biomechanics (ergonomics), progressive strengthening and functional rehabilitation.²⁵

CAUSES OF WRIST NUMBNESS AND HAND PAIN

Patients may present with wrist numbness or paraesthesia. This suggests a neurological pathology, most likely carpal tunnel syndrome or ulnar nerve compression.

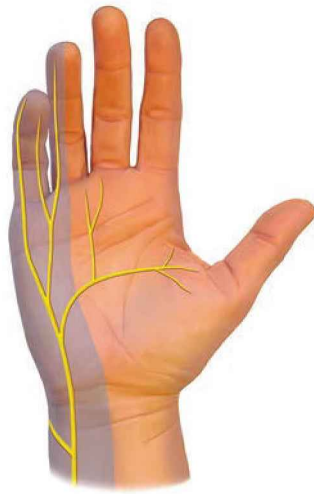
Carpal tunnel syndrome

The median nerve may be compressed as it passes through the carpal tunnel along with the flexor digitorum profundus, flexor digitorum superficialis and flexor pollicis longus tendons (Fig. 26.20). This condition is characterised by burning volar wrist pain with numbness or paraesthesia in the distribution of the median nerve (thumb, index finger, middle finger and radial side of the ring finger). Nocturnal paraesthesia is characteristic. The pain can radiate to the forearm, elbow and shoulder. Tinel's sign may be elicited by tapping over the volar aspect of the wrist (Fig. 26.20). A positive Phalen's test will reproduce symptoms in up to 60 seconds when the patient rests both elbows on a table with forearms vertical and their wrists flexed by gravity.

The most important aspects in diagnosis are the history and physical examination, but nerve conduction studies can help confirm the diagnosis and may predict how the patient will respond to surgery.⁸⁰ Patients with carpal tunnel syndrome should be checked for diabetes mellitus, as it is a known risk factor.



Figure 26.20 Median nerve compression in the carpal tunnel



(a)



(b)

Figure 26.21 Ulnar nerve compression (a) Ulnar nerve (b) Mechanism of nerve compression in cycling

Mild cases may be treated conservatively with non-steroidal anti-inflammatory drugs (NSAIDs) and splinting.^{81,82} A single corticosteroid injection may provide temporary relief,^{82,83} but persistent cases require surgical treatment.⁸⁴ Surgery may be either open or endoscopic, and systematic reviews to date show no difference between the two techniques for symptom relief.⁸⁵

Ulnar nerve compression

The ulnar nerve may be compressed at the wrist as it passes through Guyon's canal, sometimes with ulnar artery thrombosis. It is called hypothenar hammer syndrome, and it is most commonly seen in cyclists due to supporting body weight over a long duration when cycling,⁸⁶ because of poor bike fit or a failure to use several relaxed handlebar grip positions (Fig. 26.21). It also occurs in karate practitioners, and one study highlighted the risk of hand neurovascular changes in baseball players, especially catchers, from repeated trauma associated with catching a ball.⁸⁷ Within Guyon's canal, the nerve lies with the ulnar artery between the pisiform bone on the ulnar side and the hamate radially.

Symptoms include pain and paraesthesia to the little finger and ulnar side of the fourth finger. Weakness usually develops later. For cyclists, non-surgical treatment

involves splinting, NSAIDs and changes to bicycle set up. In particular, the grip position should be assessed, and moving the saddle backwards may reduce the amount of weight going through the cyclist's upper limbs. Surgical exploration of Guyon's canal may be required.

SURGERY FOR WRIST CONDITIONS

Traditional indications for surgery are a failure of conservative treatment together with an applicable surgical procedure that has proven outcome results. In many weekend athletes this holds true. However, in professional athletes with a limited career time span, the career goals impact heavily on both the indication and the timing of surgery.

In the management of intersection syndrome, aggressive use of early surgery with immediate return to elite rowing means less loss of condition and changes to technique that may impact on the sports results and hence the career of the rower. The use of more sturdy fixation than normal for a wrist fracture allows return to sport earlier and with more confidence. The risks of surgery must be discussed with the individual athlete to measure the risk-benefit ratio and to discuss the subsequent rehabilitation. (See Chapter 2—shared decision making). Table 26.4 indicates conditions where an early surgical opinion and liaison with the athlete support team are warranted.

Table 26.4 Urgent surgical referrals for wrist conditions in sports

Injury	Common sports	Management
Fractured radius	All high-impact sports	Immediate assessment of need for surgery (X-ray/specialist)
Fractured scaphoid	Contact sports/riding sports	Early diagnosis by X-ray or CT scan and early surgery best option
Extensor carpi ulnaris (ECU) dislocation or subluxation	Wrist rotation sports (especially tennis)	Early splinting but often requires surgery
Perilunate dislocation	Contact sports	Emergency joint reduction and pinning (median nerve at great risk)
Kienböck's disease (avascular necrosis of lunate)	All sports but especially golf and tennis	Splint very early but most require specialist reconstruction
Acute carpal tunnel syndrome	Any sports	Numbness must be treated urgently to avoid long-term disability
Forearm compartment syndrome (acute)	Crush injuries (high impact)	Decompression urgently

WRIST REHABILITATION

The key factor in wrist rehabilitation is the balance between the need to immobilise or restrict movement and the requirement to restore full range of movement.

Wrist splinting

Correct positioning of a wrist within a cast or splint is crucial to its potential to heal. Commercially available splints rarely fit well enough, and so custom fabrication is recommended for most conditions. If immobilisation of the fingers is not required, then the cast should end at the distal palmar crease to allow full finger flexion. Casting of non-displaced scaphoid fractures does not always require the thumb to be included, but it is preferred to do so.⁸⁸

The material used will depend on the injury, with plaster or fibreglass best for fractures, and thermoplastic more suited to tendinopathies, strains, postoperative splinting and protection during activity. The rules governing the use of splints in sport vary widely between competitions and levels, making it the treating clinician's responsibility to know what is permissible.

Post-immobilisation wrist rehabilitation

A wrist that has been casted or splinted for an extended period of time will be stiff and weakened when the cast is removed. Management of the expectations of the athlete is paramount. There is no point in restoring full wrist motion at the expense of stability and control. Oedema must be minimised and compression bandages and wraps are useful. Ongoing use of a splint may be required

overnight or in between exercise sessions to minimise pain and inflammatory responses. Its presence can also be a reminder to the patient not to do too much in the initial stages.

Joint mobilisation, stretching, active movement in all planes and graduated strengthening should begin as soon as healing permits, as well as the incorporation of functional activities such as bouncing a ball. The dart thrower's motion (mid-range radial extension to ulnar flexion) is a stable movement that can be used for early mobilisation in situations where ligamentous injury has not been ruled out.⁸⁹

Functional wrist strengthening and neuromuscular training exercises may include:

- bouncing a ball
- ball/towel stretch on a table or wall
- holding a racquet, rolling a ball on strings
- light catching (alter the ball size)
- throwing a ball at a mini-trampoline
- slosh pipe (an unstable weight created by putting water inside a plastic pipe)
- gyro-ball
- tossing ball, underhand/overhand
- TheraBand™
- wrist weights
- weight-bearing progression through putty, ball, wobble board
- putting
- curling a cricket bat
- rotation with a hammer.

REFERENCES

References for this chapter can be found at www.mhhe.com/au/CSM5e